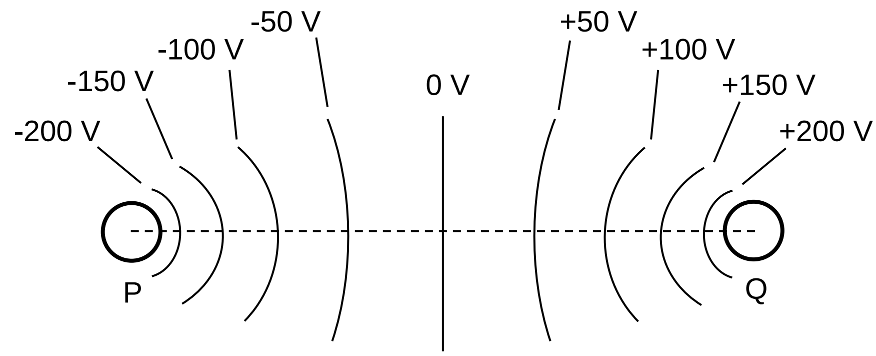
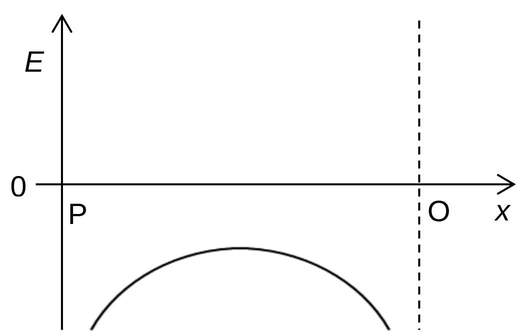
21 A charged object is placed at point P and another charged object is placed at point Q.

The diagram shows a number of solid lines along which the electric potential has a constant value.

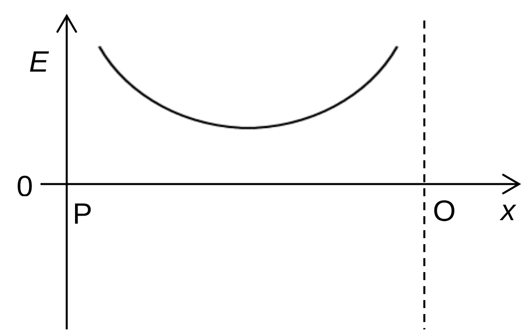


Taking vectors to the right as positive, which graph shows the variation with distance *x* along the line PQ of the electric field strength *E*?

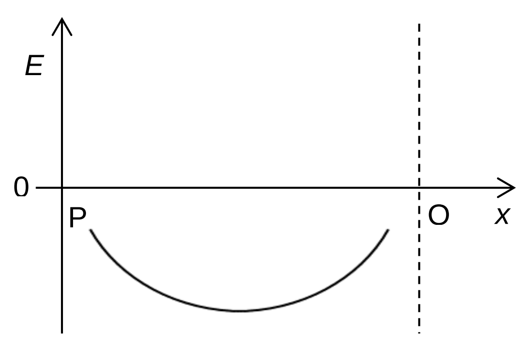
A



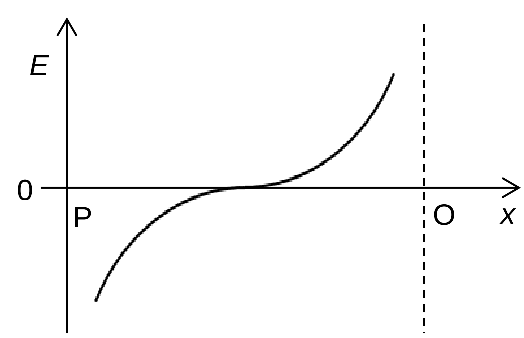
B



C



D



Answer: A

$$E = -\frac{dV}{dx}$$

Since the **electric potential V decreases all the way from Q to P**, and the **potential gradient is non-zero everywhere along PQ**,
the electric field along line PQ is **non-zero** everywhere along PQ (hence answer **cannot be D**) and,

the direction of the electric field at every point along PQ is **leftwards**, which by the sign convention is the negative direction, hence ***E* is negative everywhere along PQ** (hence **answer can only be A or C)**.

Also, **where the potential gradient is greater, *E* is greater in magnitude.** As shown in the equipotential map, **nearer P and nearer Q where the equipotential lines get closer, i.e. Δx decreases while ΔV unchanged, magnitude of E should be greater.** Hence **answer must be A.**

22	In bright light, a light-dependent resistor (LDR) has a resistance of *R*. It is connected in series